

PROGRAMMING IN HASKELL



Topic 11.1 – Semigroups and Monoids

Composability

1

e.g. to use ++ for composing a string:

```
"this" ++ " " ++ "is" ++ " " ++  
"a" ++ " " ++ "bit" ++ " " ++ "much"
```

Yes, it is a bit too much! We need a better way!

Composability

2

Composability means that you create something new by combining two like things, e.g.

```
myLast :: [a] -> a  
myLast = head . reverse
```

```
myMin :: Ord a => [a] -> a  
myMin = head . sort
```

```
myMax :: Ord a => [a] -> a  
myMax = myLast . sort
```


Composing types – Semigroups and Monoids

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A *semigroup* is a type class that has one class has only one important method you need, the `<>` operator.

You can think of `<>` as an operator for combining instances of the same type. You can trivially implement Semigroup for Integer by defining `<>` as `+`.

```
instance Semigroup Integer where  
  (<>) x y = x + y
```


Semigroups

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Note the type of `<>`

```
(<>) :: Semigroup a => a -> a -> a
```

This simple signature is the heart of the idea of composability

You can take two like things and combine them to get a new thing of the same type.

Semigroups - example

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Looking at combining colours:

- Adding Blue and yellow make green.
- Red and yellow make orange.
- Blue and red make purple.

```
data Color = Red |  
            Yellow |  
            Blue |  
            Green |  
            Purple |  
            Orange |  
            Brown deriving (Show, Eq)
```

Semigroups - example

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How we
combine
the
colours

```
instance Semigroup Color where
  (<>) Red Blue = Purple
  (<>) Blue Red = Purple
  (<>) Yellow Blue = Green
  (<>) Blue Yellow = Green
  (<>) Yellow Red = Orange
  (<>) Red Yellow = Orange
  (<>) a b = if a == b
              then a
              else Brown
```


Semigroups - associativity

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Associativity : e.g.
 $(1 + (2 + 3)) == (1 + 2) + 3$

```
instance Semigroup Color where
  (< >) Red Blue = Purple
  (< >) Blue Red = Purple
  (< >) Yellow Blue = Green
  (< >) Blue Yellow = Green
  (< >) Yellow Red = Orange
  (< >) Red Yellow = Orange
  (< >) a b | a == b = a
            | all (`elem` [Red,Blue,Purple]) [a,b] = Purple
            | all (`elem` [Blue,Yellow,Green]) [a,b] = Green
            | all (`elem` [Red,Yellow,Orange]) [a,b] = Orange
            | otherwise = Brown
```

Theoretically, all instances of semigroups should be associative but these laws are not enforceable by the Haskell compiler. Associativity should be implemented.

Semigroups -> Monoids

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A monoid is a semigroup

Monoids are similar to semigroups

we call `<>` mappend

we have mempty (identity)

we have mconcat

```
class Monoid a where
  mempty  :: a
  mappend :: a -> a -> a
  mconcat :: [a] -> a
```


Monoids

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```
GHCi> [1,2,3] ++ []  
[1,2,3]
```

```
GHCi> [1,2,3] <> []  
[1,2,3]
```

```
GHCi> [1,2,3] `mappend` mempty  
[1,2,3]
```

All three
are the
same

Monoids

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Type definition of mconcat:

```
mconcat :: Monoid a => [a] -> a
```

Takes a
list and

```
GHCi> mconcat ["does", " this", " make", " sense?"]  
"does this make sense?"
```

returns the
flattened list

Monoids

11

Definition of mconcat:

```
mconcat = foldr mappend mempty
```

```
GHCi> mconcat ["does"," this"," make"," sense?"]  
"does this make sense?"
```

```
"does" : ( "this" : ( " make" : ( " sense?": mempty ) ) )
```

++

[]

++

[]

